

Ashkan Moradi

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Education

- Jan 2025 – Present 📖 **Ph.D. student in Telecommunication, Institut national de la recherche scientifique (INRS), Montreal, Canada.**
Thesis title: *Securing the Metaverse Through User Access Control.*
- Sep 2018 – Feb 2022 📖 **M.Sc. Artificial Intelligence and Robotics, Shahid Beheshti University, Tehran, Iran.**
Thesis title: *Performance Improvement of Spoken Language Identification System via Fusion of the Acoustic and Phonetic Information based on learning methods*, (Grade: A+).
GPA: 16.23 out of 20.
- Sep 2013 – Jan 2018 📖 **B.Sc. Computer Software Engineering, Sadjad University of Technology, Mashhad, Iran.**
Thesis title: *Designing and Developing a Website about Machine Learning*, (Grade: A+).
GPA: 16.18 out of 20.

Publications

- 1 Alipour Esgandani, S., Shekofteh, Y., & **Moradi, A.** (2024). Apedm: A new voice casting system using acoustic–phonetic encoder–decoder mapping. *Multimedia Tools and Applications*, 1–25.
- 2 **Moradi, A.**, & Shekofteh, Y. (2023). Spoken language identification using a genetic-based fusion approach to combine acoustic and universal phonetic results. *Computers and Electrical Engineering*, 105, 108549.
- 3 **Moradi, A.**, Shekofteh, Y., & Zarei, S. (2021). A genetic-based fusion approach of persian and universal phonetic results for spoken language identification. In *2021 11th international conference on computer engineering and knowledge (iccke)* (pp. 175–181). IEEE.

Work Experience

Teaching Assistant

- 2020 📖 **Digital Speech Processing**, Shahid Beheshti University, Tehran, Iran.
📖 **Open Banking**, Tarbiat Modares University, Tehran, Iran.
- 2017 📖 **Artificial Intelligence**, Sadjad University of Technology, Mashhad, Iran.
📖 **Algorithms Design and Analysis**, Sadjad University of Technology, Mashhad, Iran.
- 2016 – 2017 📖 **Data Structures**, Sadjad University of Technology, Mashhad, Iran.
📖 **Operating Systems**, Sadjad University of Technology, Mashhad, Iran.
📖 **Advanced Programming**, Sadjad University of Technology, Mashhad, Iran.

Work Experience (continued)

2015  **Introduction to Programming**, Sadjad University of Technology, Mashhad, Iran.

Professional Experience

Nov 2023 – Nov 2024  **Tajan System**, .NET Developer, Tehran, Iran.

Jan 2022 – Sep 2023  **Techofen** R&D Manager, Tehran, Iran.

Mar 2020 – Oct 2023  **Techofen**, .NET Developer, Tehran, Iran.

Nov 2018 – Mar 2020  **Saman Insurance Co.**, .NET Developer, Tehran, Iran.

Projects and Research

- 2024  **Employee Performance Evaluation System**: In this project, we developed an enterprise system to evaluate employees across various departments within a company. It featured monthly, seasonal, and yearly report cards for performance comparison, along with tools like polling, peer feedback, goal tracking, and customizable evaluation templates.
- 2022  **Engine Fault Detection Using Engine Sound**: In this project, we are examining and using different acoustic features to detect engine faults in an unsupervised manner.
- 2020 – 2022  **Performance Improvement of Spoken Language Identification System via Fusion of the Acoustic and Phonetic Information based on learning methods**: This was my M.Sc. thesis project, for which I developed an acoustic-based x-vector system and a phonetic system based on the Allosaurus universal phone recognizer. Finally, I proposed two different fusion methods based on a genetic algorithm to fuse the mentioned system.
- 2020  **Sentiment Analysis on SOHA system of Shahid Beheshti University**: The goal of this project was to classify comments of SOHA systems into three groups: positive, negative, and neutral. Since these comments were not labeled, I used a lexicon-based approach to classify them.
- 2020  **Implementation of "Many Objective Cooperative Bat Searching Algorithm"**: In this project, I implemented the "Many objective cooperative bat searching algorithm" paper to optimize problems with different objective functions.
- 2019  **Credit Card Fraud Detection Using Hmm**: This project aimed to detect credit card fraudulent transactions using the Hidden Markov Model and different clustering methods.
- 2019  **Multiclass Semantic Segmentation**: The goal of this project was to implement multi-class semantic segmentation using Fully Convolutional Networks (FCN) on the CamVid dataset.
- 2018  **Constrained Clustering**: The goal of this project was to add some constraints to the K-means clustering algorithm. The project was done based on active-learning methods.
- 2017  **Speaker Gender Detection**: The project was based on some speech-related features such as MFCC, pitch period, short-time energy, and spectral centroid.

Skills

Languages  **Persian**: Native proficiency, **English**: Full professional proficiency, **French**: Elementary proficiency, **Spanish**: Elementary proficiency.

Skills (continued)

- Coding  Python, PyTorch, TensorFlow, Matlab, C#, C++, JS, .Net, LaTeX.
- AI  Speaker Profiling, Language and Speaker Recognition, Speech and Signal Processing, Speech Foundation Models, Machine/Deep Learning, Natural Language Processing.
- Other  Kaldi, Linux, Docker, Team Management, Project Management, Time Management, System Design, Full Stack Developer, Rest API.

Certifications and Training

- 2023  **Custom and Distributed Training with TensorFlow**, by Coursera.
-  **Custom Models, Layers, and Loss Functions with TensorFlow**, by Coursera.
- 2022  **DeepLearning.AI TensorFlow Developer Specialization**, by Coursera.
-  **Sequences, Time Series and Prediction**, by Coursera.
-  **Natural Language Processing in TensorFlow**, by Coursera.
-  **Machine Learning Specialization**, by Coursera.
-  **Machine Learning: Clustering & Retrieval**, by Coursera.
-  **Convolutional Neural Networks in TensorFlow**, by Coursera.
-  **Introduction to TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning**, by Coursera.
-  **Python for Deep Learning: Build Neural Networks in Python**, by Udemy.
-  **Machine Learning: Classification**, by Coursera.
-  **Machine Learning From Basic to Advanced**, by Udemy.
-  **NLP Course for Beginner**, by Udemy.
- 2018  **Machine Learning: Regression**, by Coursera.
-  **Applied Machine Learning in Python**, by Coursera.
-  **Machine Learning Foundations: A Case Study Approach**, by Coursera.
-  **Applied Plotting, Charting & Data Representation in Python**, by Coursera.
-  **Introduction to Data Science in Python**, by Coursera.

Miscellaneous

- 2019  **Best data mining project of the year award winner** at the university.
- 2016  **ACM/ICPC 2016 Technical Committee**, by Sadjad University of Technology, Mashhad, Iran.
- 2015  **ACM/ICPC 2015 Excecutive Committee**, by Sadjad University of Technology, Mashhad, Iran.